Marius P. Furter

Curriculum Vitae

University of Zurich
Institute of Mathematics
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Personal Data

Full Name: Marius Paul Furter
Date of Birth: 6 November 1994
Nationalities: Switzerland / United States of America
Languages: Native English and German speaker

Education

2022–present **PhD in Mathematics**, *University of Zurich*
Compositional dynamic modeling of biosystems
Advisor: Prof. Alberto Cattaneo

2019–2022 **BSc in Mathematics**, *University of Zurich*
Single Major

2017–2019 **MSc in Interdisciplinary Sciences**, *ETH Zurich*
Major: Biology and Chemistry
Master Thesis: *Engineering Myoglobin Towards Halogenase Activity*
Supervisor: Prof. Donald Hilvert

2012–2017 **BSc in Interdisciplinary Sciences Bio-Chem.**, *ETH Zurich*

2006–2012 **Bilingual (German/English) Matura**, *Freies Gymnasium Zürich*

Teaching

Lecture Courses

Fall 2025 **MAT740 Topology in Lean**, *University of Zurich*
This course introduces students to the Lean theorem prover and its mathematical library mathlib, by formalizing basic concepts in topology.

Spring 2023/2025 **MAT605 Logic and Foundations in Haskell**, *University of Zurich*
This course covers logic, proof theory, and foundations of math with the help of the functional programming language Haskell.

Supervised Theses

Spring 2025 **Master Thesis**, *Keerthiga Rajakumar*, Kalman Filtering of Financial Time Series

Fall 2024 **Bachelor Thesis**, *Luisa Stückelberger*, Sample-Efficient and Interpretable Bayesian Model-Based Reinforcement Learning
Thompson sampling in an environment modeling sepsis treatment.

Spring 2024 **Semester Thesis**, *Justin Bollhalder*, Categorical Constructions in the Field of Logic
Basic topos theory and its applications in logic.

Spring 2024 **Master Thesis**, *Mattia Bottoni*, Teaching with LEAN
Co-supervised with Elif Sacikara.

Fall 2023 **Semester Thesis**, *Anna Stoll-Bickel*, Relations and Relationships: A Mathematical Approach to Kinship and Marriage Systems
Mathematical modeling of kinship and marriage systems using relations and Bayesian networks.

Online Lecturing

2020–present **YouTube Channel**, *www.youtube.com/@mariusfurter*
I have published over 80 mathematics video lectures. The channel has over 7,000 subscribers and receives 1,000 hours of watch-time per month. It includes series on
- Topology
- Category Theory
- Mathematical Logic
- Graph Theory

Student Seminars

Spring 2024 **MAT676 Nonstandard Analysis**, *University of Zurich*

Fall 2022 **MAT746 Euclidean Geometry**, *University of Zurich*

Teaching Assistant

Fall 2023 **MAT701 Topology**, *University of Zurich*
Held tutorial sessions, wrote exercise sheets, graded exercises, and wrote the exam for the course.

Fall 2022 **MAT182 Analysis for Natural Scientists**, *University of Zurich*
Supervised the course forum.

Spring 2022 **MAT183 Probability and Statistics for Natural Scientists**, *University of Zurich*
Supervised the course forum.

Fall 2021 **MAT182 Analysis for Natural Scientists**, *University of Zurich*
Held tutorial sessions and graded exercise sheets.

Fall 2021 **MAT101 Programming**, *University of Zurich*
Held tutorial sessions and wrote exercise sheets on Python programming.

Spring 2021 **Applied Compositional Thinking for Engineers**, *ETH Zurich*

Volunteering

2020–present **Tutor at 'incluso LERNstudio'**, *Caritas Zurich*
Every week, I help displaced persons in vocational training with their homework.

Publications

Theses

[1] Marius Furter. Engineering Myoglobin Towards Halogenase Activity. Master's thesis, ETH Zurich, Zurich, Switzerland, March 2019.

Journal Articles

[2] Armin Handzic, Marius P. Furter, Brigitte C. Messmer, Magdalena A. Wirth, Yulia Valko, Fabienne C. Fierz, Edward A. Margolin, and Konrad P. Weber. Multivariable Prediction Model for Suspected Ocular Myasthenia Gravis: Development and Validation. *Journal of Neuro-Ophthalmology*, April 2025.

[3] Puca E, Schmitt-Koopmann C, Furter M, Murer P, Probst P, Dihr M, Bajic D, and Neri D. The targeted delivery of interleukin-12 to the carcinoembryonic antigen increases the intratumoral density of NK and CD8$^+$ T cell in an immunocompetent mouse model of colorectal cancer. *Journal of gastrointestinal oncology*, 08 2020.

Preprints

[4] Yujun Huang, Marius Furter, and Gioele Zardini. On Composable and Parametric Uncertainty in Systems Co-Design, April 2025.

[5] Marius Furter, Yujun Huang, and Gioele Zardini. Composable Uncertainty in Symmetric Monoidal Categories for Design Problems, March 2025.

[6] Marius Furter. Modeling Choice in Co-Design. Student Paper, ETH Zurich, Zurich, July 2021.

Posters

[7] Marius Furter. Probabilistic Signaling Networks: Mechanistic Modeling in Markov Categories, July 2023. Presented at Applied Category Theory 2023.

Invited Talks

June 2025 **Composable Uncertainty in Symmetric Monoidal Categories for Design Problems**, *Invited talk at* Applied Category Theory *conference.*

Grants

Fall 2024 **GRC Travel Grant**, *Graduate Campus*, University of Zurich
Travel grant to support my research stay at MIT.

2022–2026 **PhD Excellence Program Scholarship**, *Digital Society Initiative*, University of Zurich
Scholarship for my PhD project proposal *'Compositional dynamic modeling of biosystems'*. Participation in the PhD Excellence program included taking 12 ECTS of courses that focused on the societal effects of digitalization. The program fostered interaction between participants from various backgrounds including political sciences, media sciences, computer sciences, philosophy and law.

Visiting Positions

Fall 2024 **Visiting Student**, *Prof. Gioele Zardini*, Massachusetts Institute of Technology
During this four-month research stay, I worked on integrating probabilistic uncertainty into the monotone co-design framework.

Computer skills

	Level	Skill	Active Years	Comment
Programming	■■■□□	Julia	4	*Probabilistic programming in Turing and Gen. Developed PPL for SMC sampling using meta-programming. ODE modeling with ModelingToolkit and Catalyst. Data science with DataFrames, JuliaStats and Makie.*
	■■■□□	R	6	*Working with tidyverse. Bayesian statistics with RStan. Web applications in shiny.*
	■■■□□	Python	7	*MCMC and VI in NumPyro, Deep learning in PyTorch.*
	■■□□□	Haskell	4	
	■□□□□	Lean	3	*Automated theorem proving.*
Web	■■□□□	HTML	2	
	■■□□□	JavaScript	2	

	■■□□□	React	2	*Built an interactive Bayesian network editor.*
Typesetting	■■■□□	$\LaTeX$	7	*Used for most writing.*
Media	■■■□□	Premiere Pro		*Edited over 100 hours of video footage.*
	■■□□□	Illustrator		*Logos and thumbnails.*
	■■□□□	Photoshop		
Office	■■■■□	Microsoft Office		*Word, Excel, PowerPoint.*
	■■■□□	Apple		*Pages, Keynote, Numbers.*

■□□□□	basic knowledge	■■■□□	extensive project experience
■■□□□	intermediate knowledge with some project experience	■■■■□	deepened expert knowledge
		■■■■■	expert / specialist

Laboratory skills

I am well-versed in standard laboratory techniques in the fields of molecular biology, biochemistry and organic chemistry.

Last updated: October 1, 2025.